Testing Spring Boot Applications Demystified

A Hero's Journey Through the Spring Boot Testing Labyrinth

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About Philip

- Self-employed developer from Herzogenaurach, now living in Erlangen, Germany (Bavaria) 🍻
- Blogging & content creation with a focus on testing Java and specifically Spring Boot applications 🍃
- Founder of PragmaTech GmbH - **Enabling Developers to Frequently Deliver** Software with **More Confidence** 🚤

Participate During the Talk

Go to menti.com and use the code **3982 6427** to **anonymously** submit answers for the quizzes and add your questions during the talk.

Please answer the **first three questions**:

- Who wrote your tests this week?
- How would you rate your Spring Boot testing knowledge?
- How confident are you deploying on a Friday afternoon?

Why Test Software?

In 2026, most of our code is not written by us.

Frog told the LLM to write tests for all the code it wrote.

“There,” he said. “Now the model can’t write broken or buggy code.”

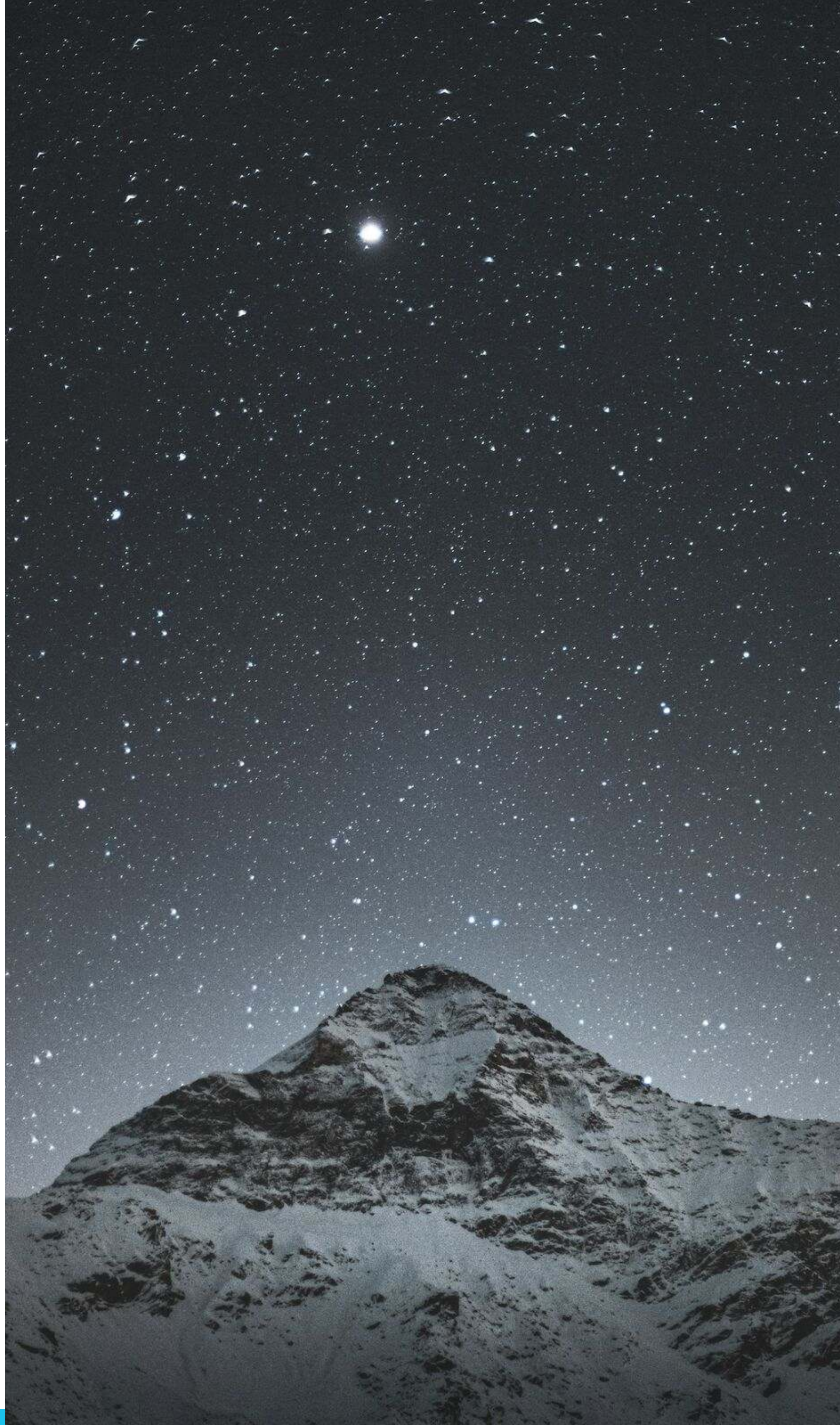
“But the model can write broken or buggy tests,” said Toad.

“That is true,” said Frog.

My Overall Northstar for Engineering Excellence

Imagine seeing this pull request on a Friday afternoon:

How **confident** are you to merge this major Spring Boot upgrade and **deploy** it to production once the pipeline turns green?



Good tests don’t just catch bugs - they give you fast feedback and confident deployments.

How do we get there for our Spring Boot projects?

The Hero's Journey aka. Our Agenda

Goals For This Talk

- **Provide a clear mental map** for choosing between unit, slice, and integration tests
- Get **better judgment** for AI-written tests
- Understand the **testing concepts** to better reason about a test strategy or test failures
- **Build confidence** into your development work to **ship fearlessly**

Act 1: The Grand Entrance

Testing Spring Boot applications can feel like entering a labyrinth blindfolded:

- Copying test configuration from AI/StackOverflow hoping it works
- The paradox of choice: `@SpringBootTest`, `@WebMvcTest`, `@DataJpaTest`, `@MockBean`, etc.
- Struggling with Spring `ApplicationContext` creation during tests
- Uncertainty about what to test and how to test it effectively

QUEST 1

The Unit Testing Guardian

The Swift Gatekeeper - Blocks Those Who Overcomplicate

Our Foundation: Spring Boot Starter Test

- The "Testing Swiss Army Knife"

```xml
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
```

- Batteries-included for testing by transitively including popular testing libraries
- Out-of-the-box dependency management to ensure compatibility

```
./mvnw dependency:tree
[INFO] ...
[INFO] +- org.springframework.boot:spring-boot-starter-test:jar:4.0.2:test
[INFO] |  +- org.springframework.boot:spring-boot-test:jar:4.0.2:test
[INFO] |  +- org.springframework.boot:spring-boot-test-autoconfigure:jar:4.0.2:test
[INFO] |  +- com.jayway.jsonpath:json-path:jar:2.10.0:test
[INFO] |  |  \- org.slf4j:slf4j-api:jar:2.0.17:compile
[INFO] |  +- jakarta.xml.bind:jakarta.xml.bind-api:jar:4.0.4:test
[INFO] |  |  \- jakarta.activation:jakarta.activation-api:jar:2.1.4:test
[INFO] |  +- net.minidev:json-smart:jar:2.6.0:test
[INFO] |  |  \- net.minidev:accessors-smart:jar:2.6.0:test
[INFO] |  |     \- org.ow2.asm:asm:jar:9.7.1:test
[INFO] |  +- org.assertj:assertj-core:jar:3.27.6:test
[INFO] |  |  \- net.bytebuddy:byte-buddy:jar:1.17.8:test
[INFO] |  +- org.awaitility:awaitility:jar:4.3.0:test
[INFO] |  +- org.hamcrest:hamcrest:jar:3.0:test
[INFO] |  +- org.junit.jupiter:junit-jupiter:jar:6.0.2:test
[INFO] |  |  +- org.junit.jupiter:junit-jupiter-api:jar:6.0.2:test
[INFO] |  |  |  +- org.opentest4j:opentest4j:jar:1.3.0:test
[INFO] |  |  |  +- org.junit.platform:junit-platform-commons:jar:6.0.2:test
[INFO] |  |  |  \- org.apiguardian:apiguardian-api:jar:1.1.2:test
[INFO] |  |  +- org.junit.jupiter:junit-jupiter-params:jar:6.0.2:test
[INFO] |  |  \- org.junit.jupiter:junit-jupiter-engine:jar:6.0.2:test
[INFO] |  |     \- org.junit.platform:junit-platform-engine:jar:6.0.2:test
[INFO] |  +- org.mockito:mockito-core:jar:5.5.0:test
[INFO] |  |  +- net.bytebuddy:byte-buddy-agent:jar:1.17.8:test
[INFO] |  |  \- org.objenesis:objenesis:jar:3.3:test
[INFO] |  +- org.mockito:mockito-junit-jupiter:jar:5.5.0:test
[INFO] |  +- org.skyscreamer:jsonassert:jar:1.5.3:test
[INFO] |  |  \- com.vaadin.external.google:android-json:jar:0.0.20131108.vaadin1:test
[INFO] |  +- org.springframework:spring-core:jar:7.0.3:compile
[INFO] |  |  +- commons-logging:commons-logging:jar:1.3.5:compile
[INFO] |  |  \- org.jspecify:jspecify:jar:1.0.0:compile
[INFO] |  +- org.springframework:spring-test:jar:7.0.3:test
[INFO] |  \- org.xmlunit:xmlunit-core:jar:2.10.4:test
```

What's Inside the Testing Swiss Army Knife?

- **JUnit**: Java's de-facto standard testing framework and foundation.
- **Mockito**: Creating mock objects to simulate dependencies and verify interactions.
- **AssertJ**: Provides fluent, chainable, and readable assertions.
- **Hamcrest**: Offers flexible matchers for creating custom assertions.
- **JSONAssert**: Compares JSON strings with flexible matching options.
- **JsonPath**: Extracts and queries data from JSON similar to XPath.
- **XMLUnit**: Compares and validates XML documents.
- **Awaitility**: Handles asynchronous testing with fluent conditions.

Unit Testing Java/Spring Boot Applications 101

- **Core Concept**: Test individual components in isolation from dependencies - one unit of work at a time.
- **Confidence Gained**: Fast, high-volume verification that the smallest building blocks behave correctly under various conditions.
- **Pitfall**: Poor class design leads to untestable god classes. Good tests start with good design.
- **Tools**: JUnit, Mockito, AssertJ (or Spock, TestNG, Hamcrest).

Unit Testing has Limits

Consider this sample REST controller, what could we verify with a unit test?

```java
@RestController
@RequestMapping("/api/customers")
public class CustomerController {

  private final CustomerService customerService;

  public CustomerController(CustomerService customerService) {
    this.customerService = customerService;
  }

  @PostMapping
  public ResponseEntity<Void> createNewCustomer(@Validated CustomerCreationRequest payload, UriComponentsBuilder uriBuilder) {

    String customerId = customerService.createNewCustomer(payload.firstName());

    UriComponents uriComponents = uriBuilder
      .path("/api/customers/{id}")
      .buildAndExpand(customerId);

    return ResponseEntity.created(uriComponents.toUri()).build();
  }
}
```

```java
@ExtendWith(MockitoExtension.class)
class CustomerControllerUnitTests {

  @Mock
  private CustomerService customerService;

  @InjectMocks
  private CustomerController customerController;

  @Test
  void shouldCreateCustomerWhenPayloadRequestIsValid() {
    when(customerService.createNewCustomer(anyString()))
      .thenReturn("42");

    ResponseEntity<Void> result = customerController.createNewCustomer(
      new CustomerCreationRequest("Java", "Duke", "duke@spring.io"),
      UriComponentsBuilder.newInstance()
    );

    assertThat(result.getStatusCode().value())
      .isEqualTo(201);
    assertThat(result.getHeaders().getLocation().toString())
      .isEqualTo("/api/customers/42");
  }
}
```

Things We Can't Cover With a Unit Test #0

Request Mapping: Does HTTP GET `/api/customers/{id}` actually resolve to our desired method?

```java
@Test
void shouldCreateCustomerWhenPayloadRequestIsValid() {

  // ...

  ResponseEntity<Void> result = customerController.createNewCustomer(
    new CustomerCreationRequest("Java", "Duke", "duke@spring.io"),
    UriComponentsBuilder.newInstance()
  );
}
```

Things We Can't Cover With a Unit Test #1

Validation: Will an incomplete request body result in a 400 bad request or return an accidental 201?

```java
@Test
void shouldCreateCustomerWhenPayloadRequestIsValid() {

  // ...

  ResponseEntity<Void> result = customerController.createNewCustomer(
    new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
    UriComponentsBuilder.newInstance()
  );
}
```

Things We Can’t Cover With a Unit Test #2

Serialization: Are we JSON objects serialized and deserialized correctly?

```java
ResponseEntity<Void> result = customerController.createNewCustomer(
  new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
  UriComponentsBuilder.newInstance());
```

```json
{
  "first-name": "Java",
  "last_Name": "Duke",
  "email": "duke@spring.io"
}
```

Things We Can't Cover With a Unit Test #3

Security: Are we Spring Security configuration and other authorization checks enforced?

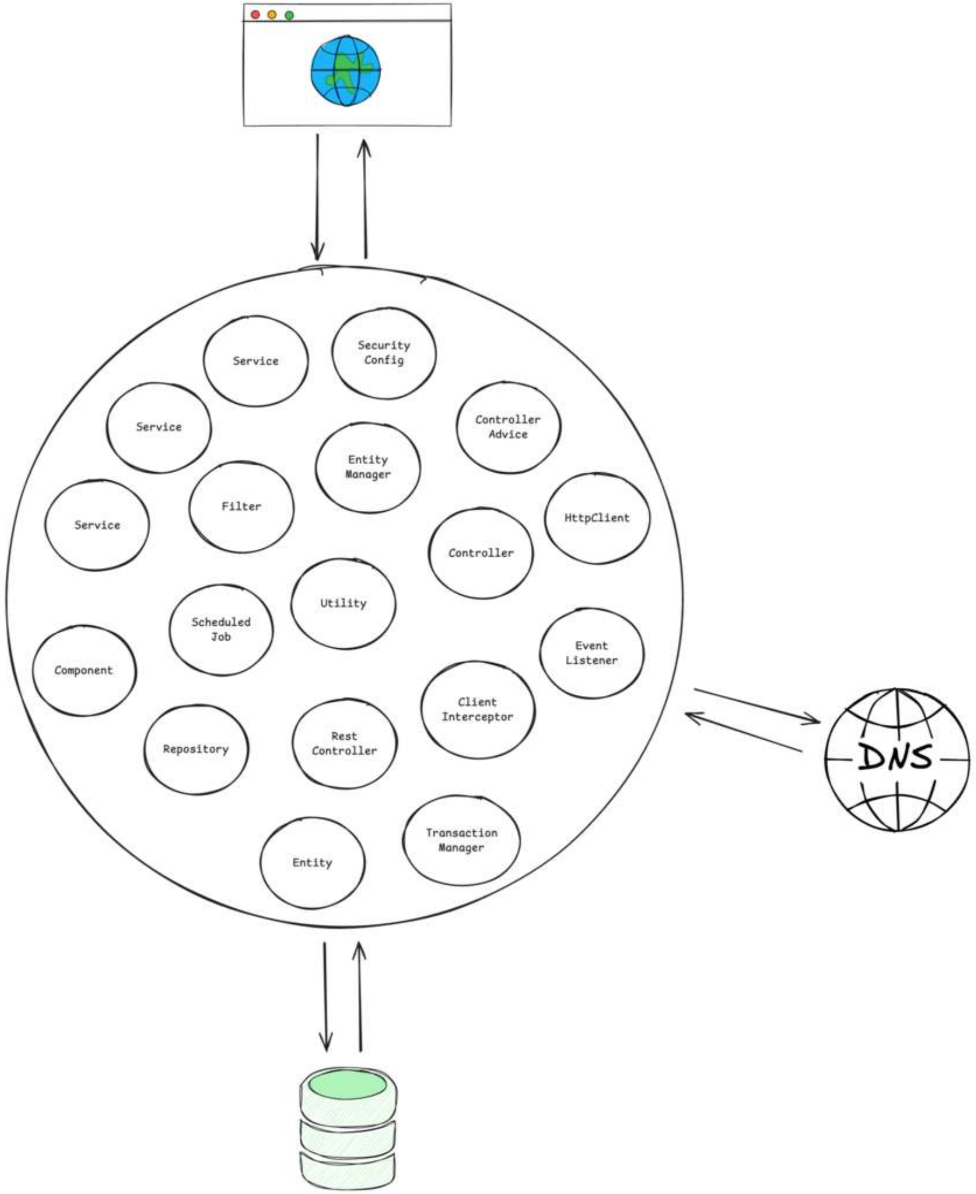
```java
@Test
void shouldCreateCustomerWhenPayloadRequestIsValid() {

  // ...

  ResponseEntity<Void> result = customerController.createNewCustomer(
    new CustomerCreationRequest("Java", "Duke", "NOT_AN_EMAIL"),
    UriComponentsBuilder.newInstance()
  );
}
```

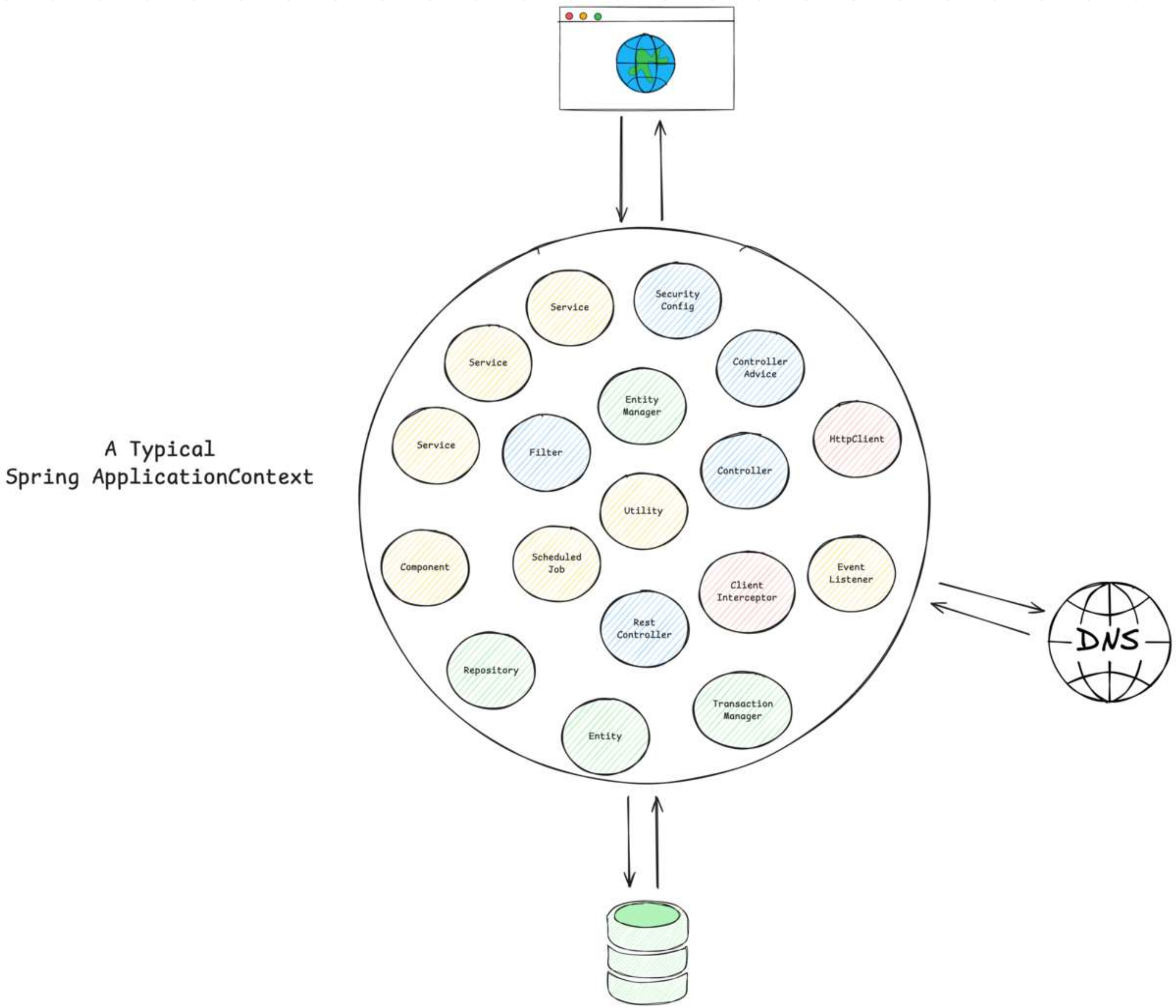
QUEST 2

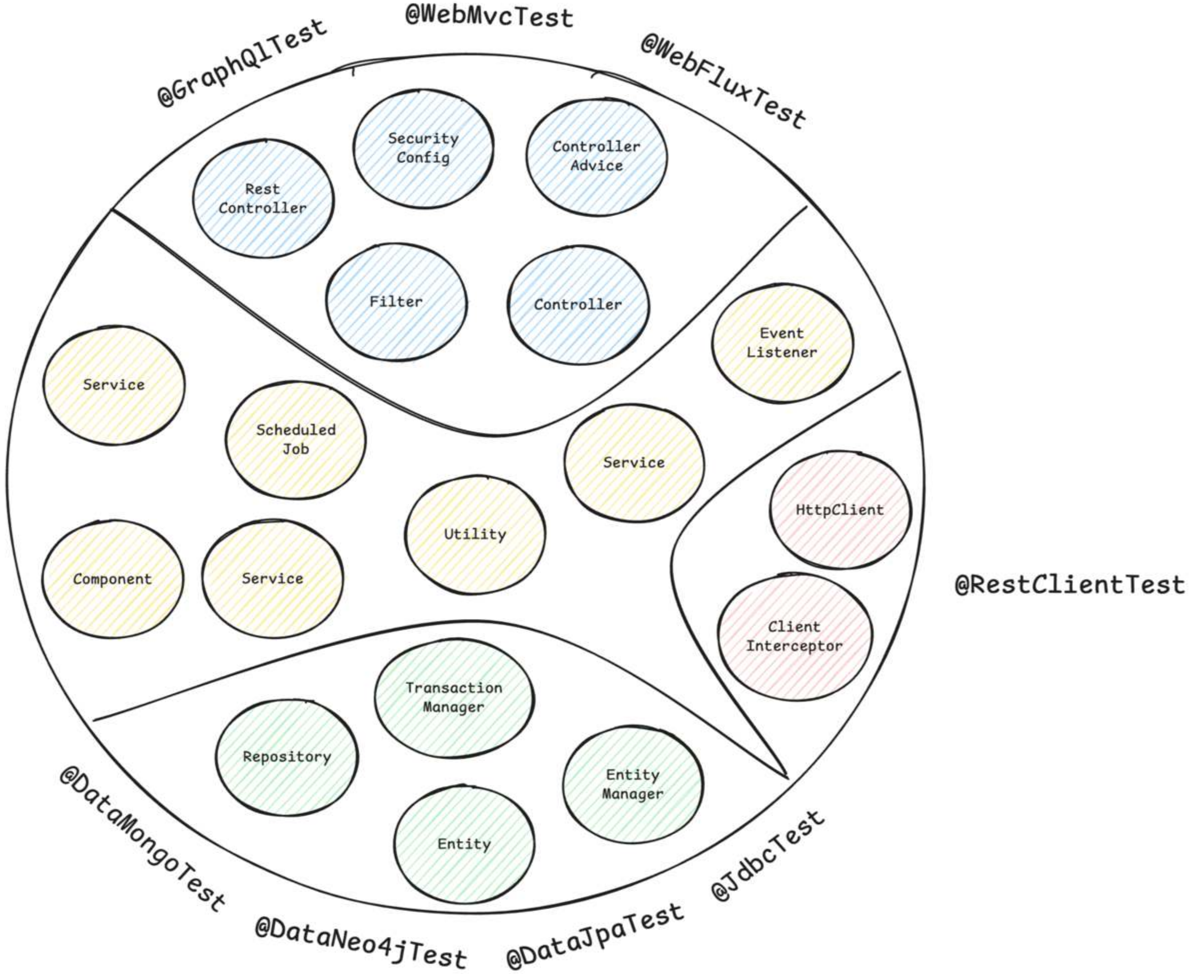
The Slice Testing Hydra

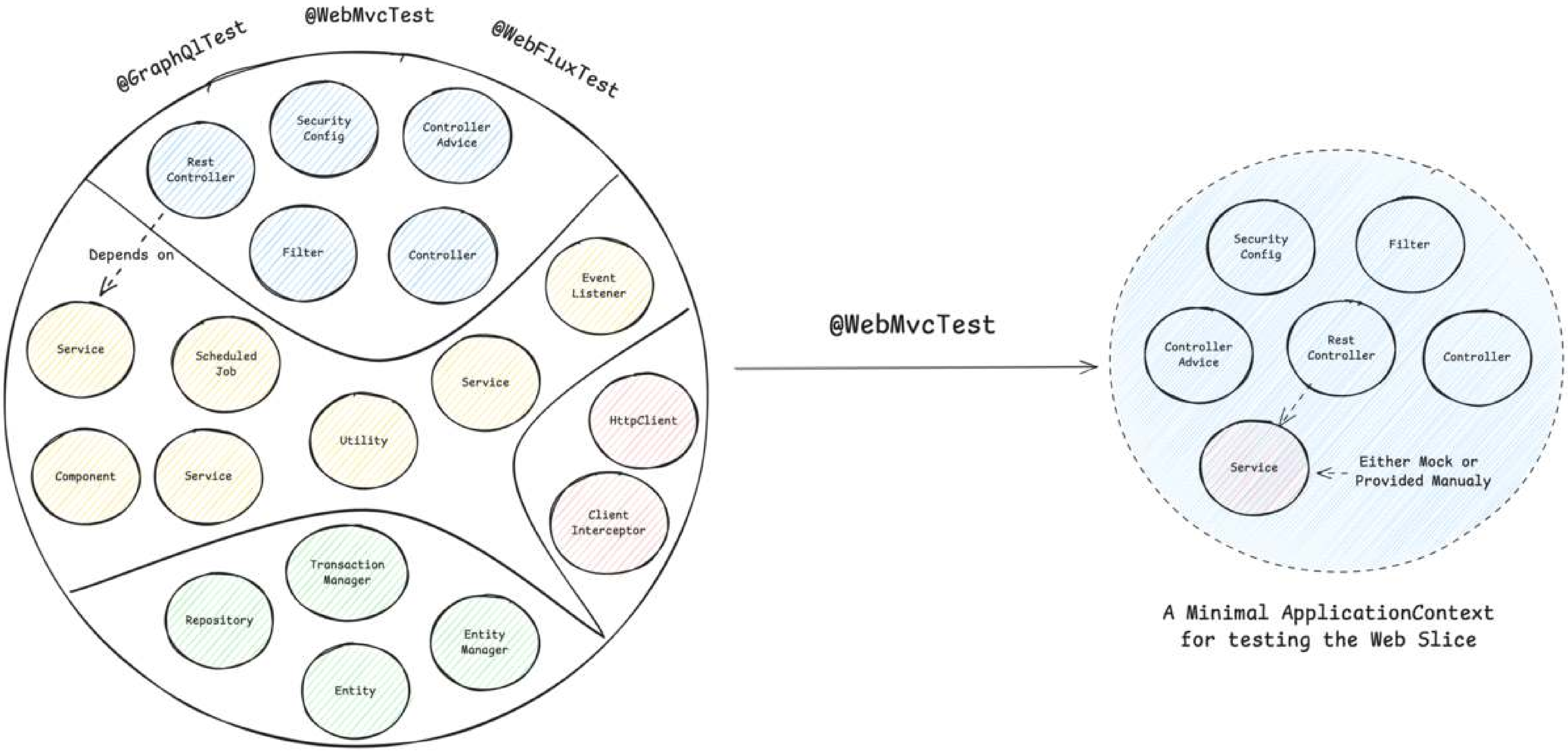
Multiple Heads, Each Guarding a Layer

A Typical
Spring ApplicationContext
with different
Spring Beans




A Typical
Spring ApplicationContext
Service
Security Config
Service
Controller Advice
Entity Manager
Service
Filter
HttpClient
Controller
Utility
Component
Scheduled Job
Event Listener
Client Interceptor
Rest Controller
Repository
Transaction Manager
Entity
DNS


@GraphQlTest
@WebMvcTest
@WebFluxTest
Security Config
Controller Advice
Rest Controller
Filter
Controller
Event Listener
Service
Scheduled Job
Service
Utility
HttpClient
Component
Service
@RestClientTest
Client Interceptor
Transaction Manager
Repository
Entity Manager
Entity
@DataMongoTest
@DataNeo4jTest
@DataJpaTest
@JdbcTest


@GraphQlTest
@WebMvcTest
@WebFluxTest
Security Config
Controller Advice
Rest Controller
Filter
Controller
Depends on
Event Listener
Service
Scheduled Job
Service
Utility
HttpClient
Component
Service
Client Interceptor
Transaction Manager
Repository
Entity Manager
Entity
@WebMvcTest
Security Config
Filter
Controller Advice
Rest Controller
Controller
Service
Either Mock or Provided Manualy
A Minimal ApplicationContext for testing the Web Slice

Spring Boot Test Slice Example: `@WebMvcTest`

```java
@WebMvcTest(CustomerController.class)
@Import(SecurityConfig.class)
class CustomerControllerTest {

  @Autowired
  private MockMvc mockMvc;

  @MockitoBean
  private CustomerService customerService;

  @Test
  @WithMockUser
  void shouldReturnLocationOfNewlyCreatedCustomer() throws Exception {
    // ...
  }
}
```

Common Test Slices

- `@WebMvcTest` / `@WebFluxTest` - Controller layer
- `@DataJpaTest` / `@JdbcTest` - Persistence layer
- `@JsonTest` - JSON serialization/deserialization
- `@RestClientTest` - RestTemplate testing
- etc.

@DataCouchbaseTest

@DataLdapTest

@RestClientTest

@JsonTest

@JooqTest

@WebFluxTest

@DataRedisTest

@WebMvcTest

@DataMongoTest

@DataCassandraTest

@GraphQlTest

@DataNeo4jTest

@WriteYourOwn*

@DataJpaTest

@SqsTest

@JdbcTest

@DataElasticsearchTest

Sliced Testing Spring Boot Applications 101

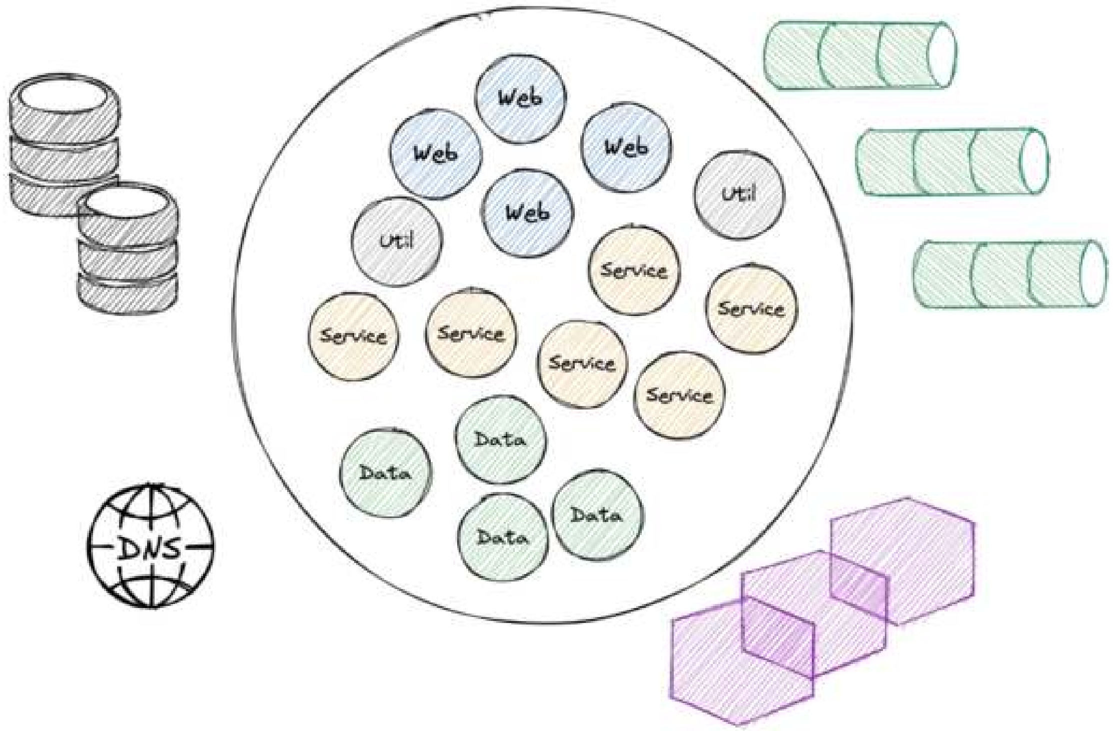
- **Core Concept**: Test a specific "slice" or layer of your application by loading a minimal, relevant part of the Spring `ApplicationContext`.
- **Confidence Gained**: Helps validate parts of your application where pure unit testing is insufficient, like the web, messaging, or data layer.
- **Prominent Examples:** Web layer (`@WebMvcTest`) and database layer (`@DataJpaTest`)
- **Pitfalls**: Requires careful configuration to ensure only the necessary slice of the context is loaded.
- **Tools**: JUnit, Mockito, Spring Test, Spring Boot, Testcontainers

QUEST 3

The Integration Testing Dragon

Guards the Full Treasure - but Demands Patience

```java
@SpringBootTest
class ApplicationTest {

  @Test
  void contextLoads() {
  }

}
```



Challenges when Starting the Entire `ApplicationContext`

- **Problem #1**: How to Ensure Surrounding Infrastructure (e.g. database, queues, etc.) is Present?
- **Problem #2**: How to Interact with our Application for Integration Tests?
- **Problem #3**: How to Keep our Build Time at a reasonable Duration?

There's Even More...

- **Problem #4**: How to handle HTTP communication from our application to remote services?
- **Problem #5**: How to provide test data and maintain a clean state between tests?
- **Problem #6**: How to handle authentication and security contexts during tests?

Provide External Infrastructure with Testcontainers (Problem #1)

Running infrastructure components (databases, message brokers, etc.) in Docker containers for our tests becomes a breeze with Testcontainers:

```java
@Container
@ServiceConnection
static PostgreSQLContainer postgres = new PostgreSQLContainer("postgres:16-alpine")
  .withDatabaseName("testdb")
  .withUsername("test")
  .withPassword("test")
  .withInitScript("init-postgres.sql");
```

This gives us an ephemeral PostgreSQL database for our tests:

```
$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS                                             NAMES
a958ee2887c6   postgres:16-alpine             "docker-entrypoint.s…"   10 seconds ago   Up 9 seconds    0.0.0.0:32776->5432/tcp, [::]:32776->5432/tcp     affectionate_cannon
ad0f804068dc   testcontainers/ryuk:0.12.0     "/bin/ryuk"              10 seconds ago   Up 9 seconds    0.0.0.0:32775->8080/tcp, [::]:32775->8080/tcp     testcontainers-ryuk-1f9f76a6-46d4-4e19-85c1-e8364da12804
```

How to Interact with our Application for Integration Tests?

Option #1:

```java
@SpringBootTest
// which is @SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.MOCK)
@AutoConfigureMockMvc
class ApplicationMockWebIT {

  // @LocalServerPort
  // private int port; <-- this would fail the test, there is no local port occupied

  @Test
  @WithMockUser
  void givenCustomersThenReturnListForAuthenticatedUser(@Autowired MockMvc mockMvc) throws Exception {
    mockMvc
      .perform(get("/api/customers")
        .header(ACCEPT, APPLICATION_JSON))
      .andExpect(status().is(200))
      .andExpect(content().contentType(APPLICATION_JSON))
      .andExpect(jsonPath("$.size()", is(1)));
  }
}
```

How to Interact with our Application for Integration Tests?

Option #2:

```java
@AutoConfigureWebTestClient // required since Spring Boot 4
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.RANDOM_PORT)
class ApplicationServletContainerIT {

  @Autowired
  WebTestClient webTestClient;

  @Test
  void contextLoads() {
    webTestClient
      .get()
      .uri("/api/customers")
      .header("Authorization", "Basic " + Base64.getEncoder().encodeToString("user:dummy".getBytes()))
      .exchange()
      .expectStatus()
      .isOk();
  }
}
```

Integration Testing Spring Boot Applications 101

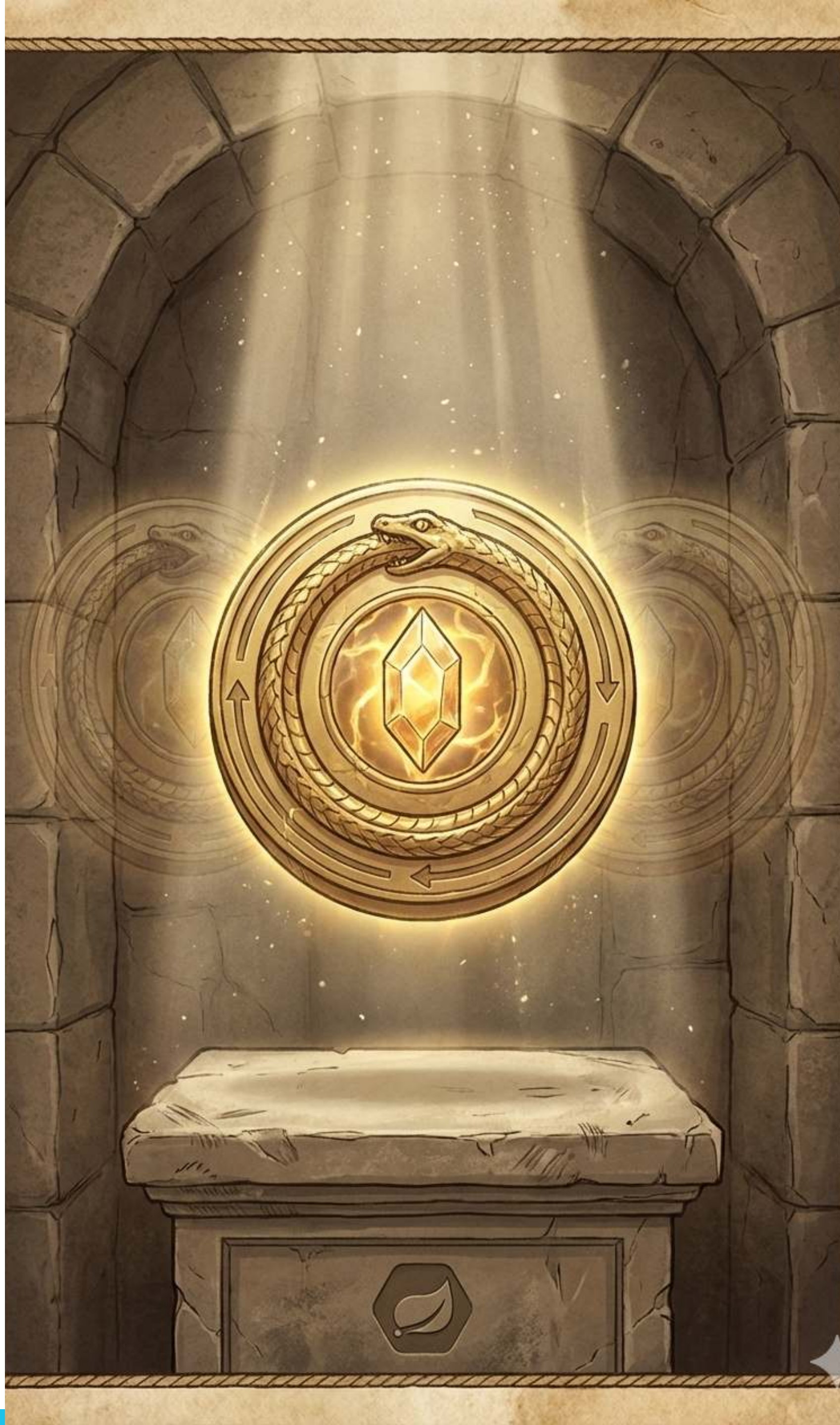
- **Core Concept**: Start the entire Spring application context, often on a random local port, and test the application through its external interfaces (e.g., REST API).
- **Confidence Gained**: Validates the integration of all internal components working together as a complete application.
- **Best Practices**: Use `@SpringBootTest` to run the app on a local port.
- **Pitfalls**: Slower to run than unit or sliced tests. Managing the lifecycle of dependent services can be complex.
- **Tools**: JUnit, Mockito, Spring Test, Spring Boot, Testcontainers, WireMock (for mocking external HTTP services), Selenium (for browser-based UI testing)

... but what about Problem #3: How to Keep our Build Time at a reasonable Duration?

QUEST ITEM 1

The Caching Amulet

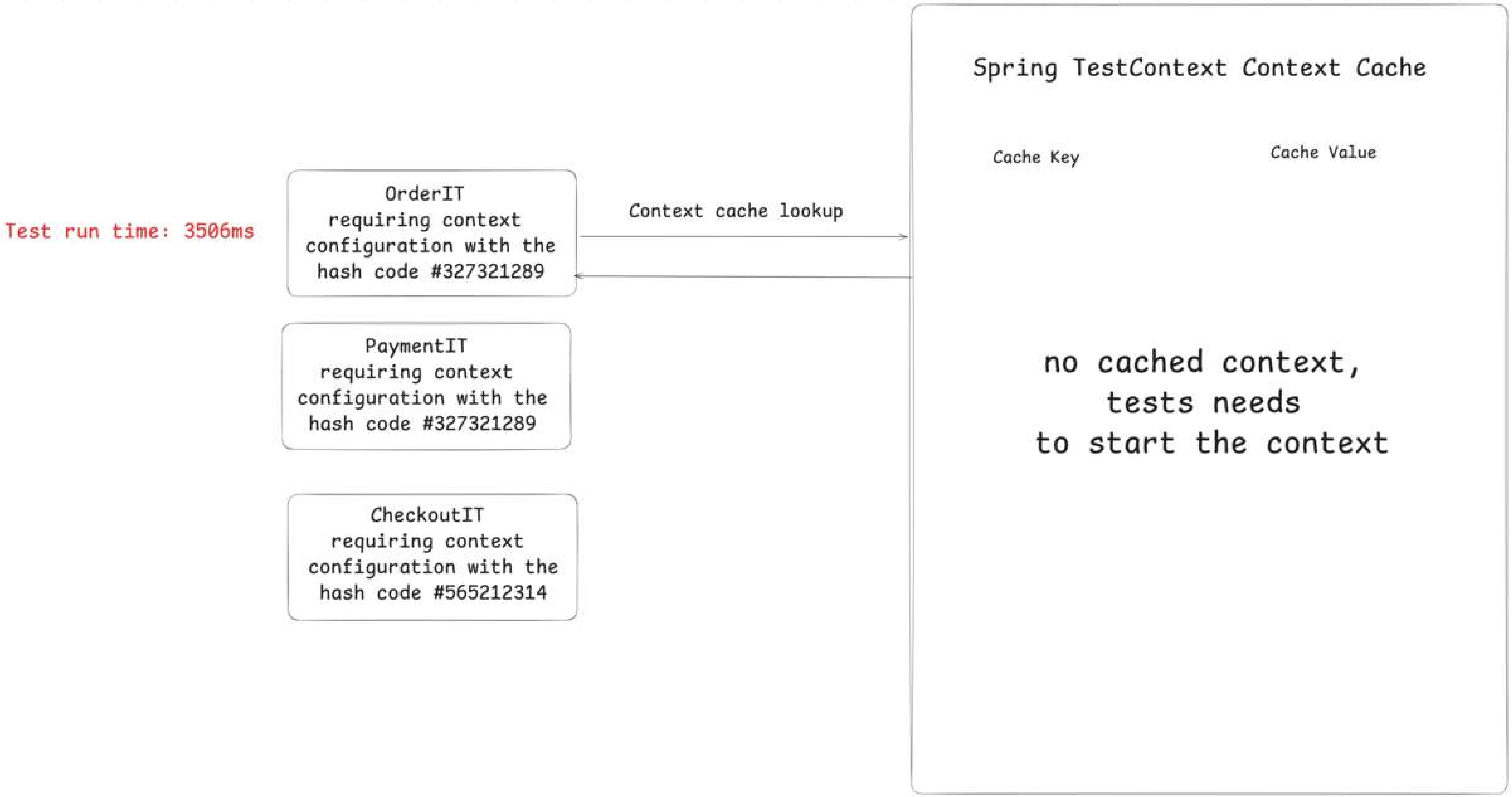
Helps You Reuse What You Already Built

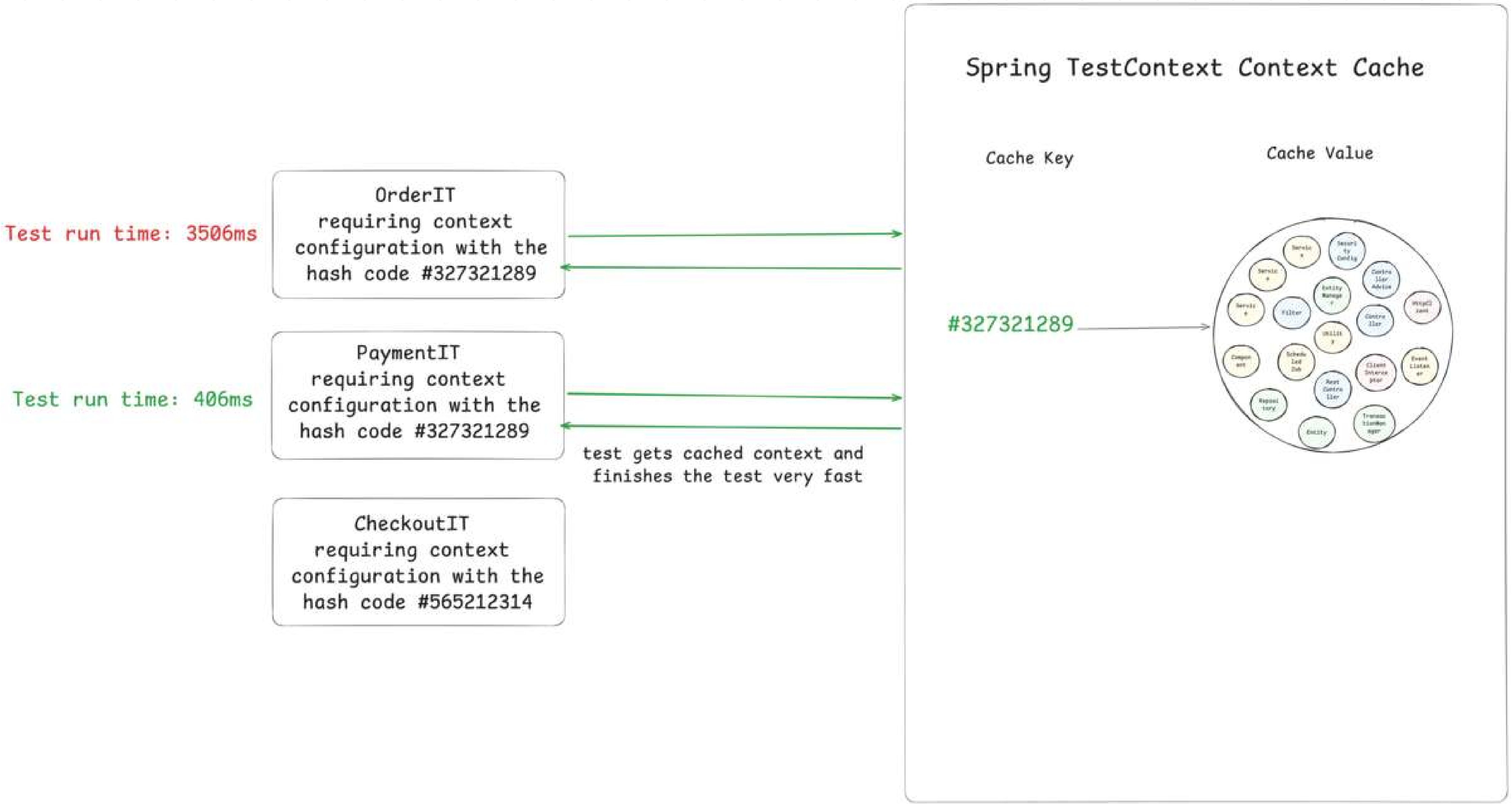


Integration Testing - The Need for Speed

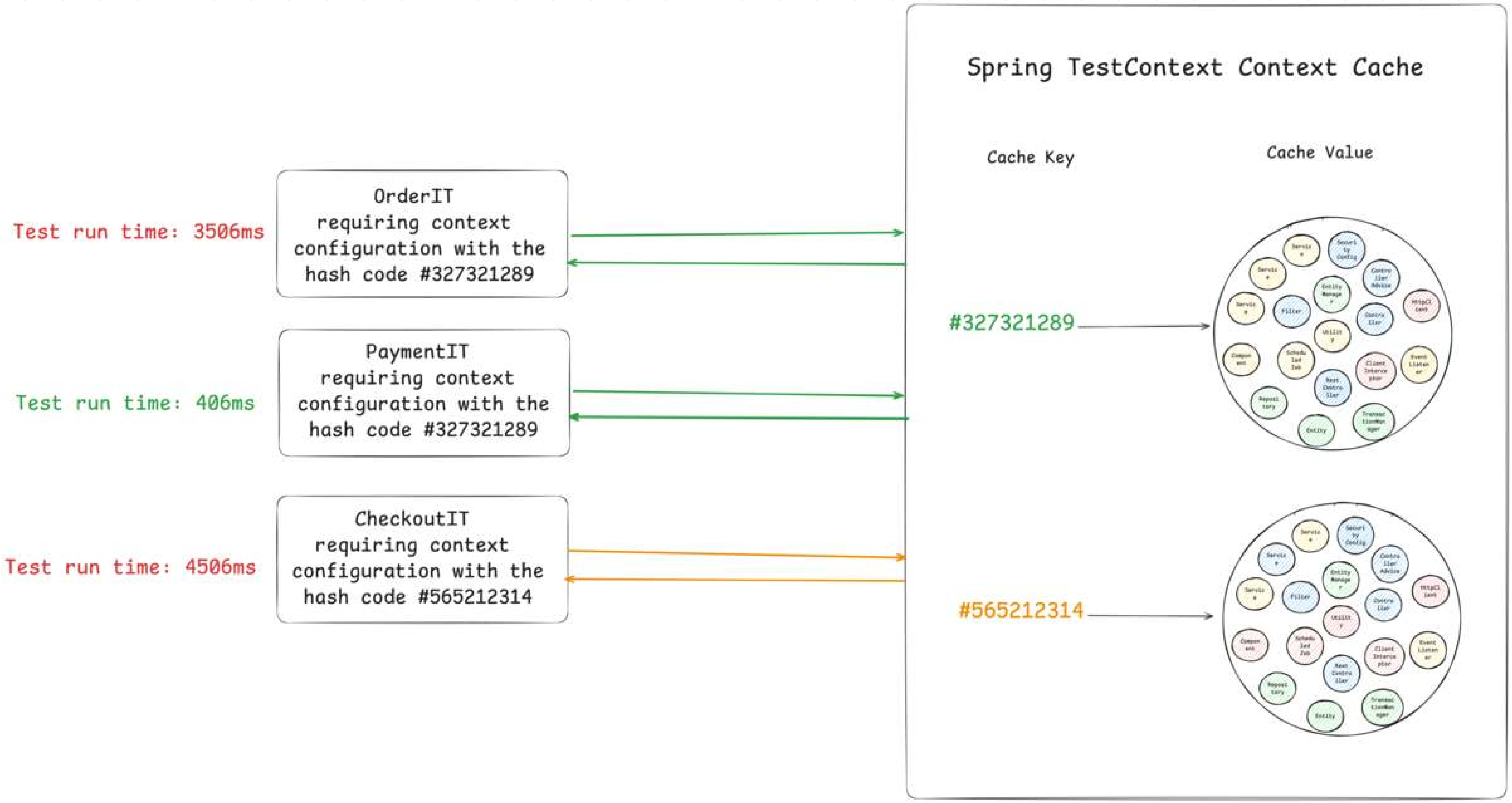
- **The Problem:** Integration tests require a started & initialized Spring `ApplicationContext`, which slows down the build
- **The Solution:** Spring Test `TestContext` Caching - stores an already started Spring `ApplicationContext` for reuse
- This feature is part of Spring Test (included in every Spring Boot project via `spring-boot-starter-test`)

Example of speed improvement:


Test run time: 3506ms
OrderIT
requiring context
configuration with the
hash code #327321289
PaymentIT
requiring context
configuration with the
hash code #327321289
CheckoutIT
requiring context
configuration with the
hash code #565212314
Context cache lookup
Spring TestContext Context Cache
Cache Key
Cache Value
no cached context,
tests needs
to start the context



Spring TestContext Context Cache
Cache Key
Cache Value
Test run time: 3506ms
OrderIT requiring context configuration with the hash code #327321289
#327321289
Test run time: 406ms
PaymentIT requiring context configuration with the hash code #327321289
test gets cached context and finishes the test very fast
CheckoutIT requiring context configuration with the hash code #565212314


Spring TestContext Context Cache
Cache Key
Cache Value
Test run time: 3506ms
OrderIT requiring context configuration with the hash code #327321289
Test run time: 406ms
PaymentIT requiring context configuration with the hash code #327321289
Test run time: 4506ms
CheckoutIT requiring context configuration with the hash code #565212314
#327321289
#565212314

How the Cache Key is Built

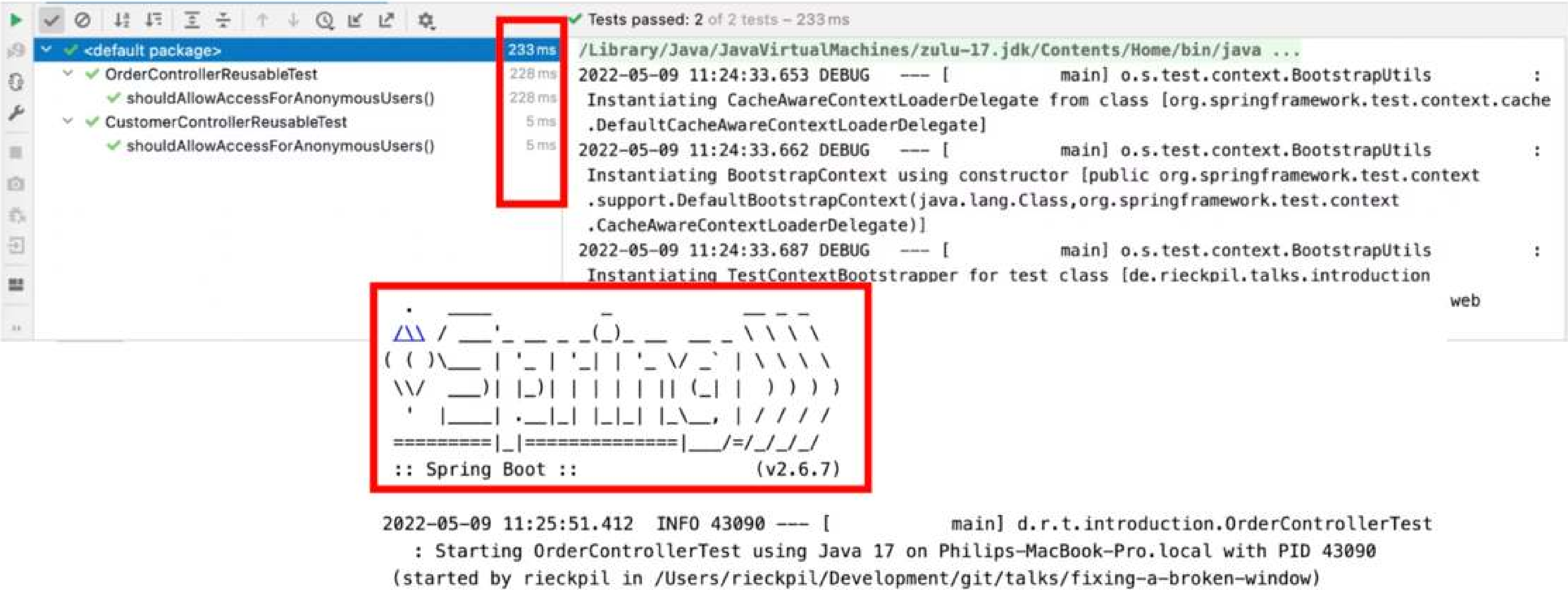
```java
// DefaultContextCache.java
private final Map<MergedContextConfiguration, ApplicationContext> contextMap =
  Collections.synchronizedMap(new LinkedHashMap<>(32, 0.75f, true));
```

The following information is part of the Cache Key (`MergedContextConfiguration`):

- activeProfiles (`@ActiveProfiles`)
- contextInitializersClasses (`@ContextConfiguration`)
- propertySourceLocations (`@TestPropertySource`)
- propertySourceProperties (`@TestPropertySource`)
- contextCustomizer (`@MockitoBean` , `@MockBean` , `@DynamicPropertySource` , ...)
- etc.

```
Test class
    │
    ▼
MergedContextConfiguration(
  testClass, locations, classes,
  activeProfiles, propertyValues,
  contextInitializers, contextCustomizers   ← every @MockitoBean lands here
  ... etc.
)
    │
    ▼  hashCode() / equals()

Cache hit? → reuse context ✅
Cache miss? → start new context and store it 🆕
```

Detect Context Restarts - Visually



Detect Context Restarts - with Logs

```
<logger name="org.springframework.test.context" level="TRACE" />
```

```
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] org.springframework.test.context.cache    : Spring test
 ApplicationContext cache statistics: [DefaultContextCache@68e62b3b size = 1, maxSize = 32, parentContextCount = 0,
 hitCount = 11, missCount = 1]
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] tractDirtiesContextTestExecutionListener : After test class:
 context [DefaultTestContext@325bb9a6 testClass = ApplicationIT, testInstance = [null], testMethod = [null], testException
  = [null], mergedContextConfiguration = [WebMergedContextConfiguration@1d12b024 testClass = ApplicationIT, locations =
 '{}', classes = '{class de.rieckpil.talks.Application}', contextInitializerClasses = '[]', activeProfiles = '{}',
 propertySourceLocations = '{}', propertySourceProperties = '{org.springframework.boot.test.context
 .SpringBootTestContextBootstrapper=true, server.port=0}', contextCustomizers = set[org.springframework.boot.test.context
 .filter.ExcludeFilterContextCustomizer@5215cd9a, org.springframework.boot.test.json
 .DuplicateJsonObjectContextCustomizerFactory$DuplicateJsonObjectContextCustomizer@9257031, org.springframework.boot.test
```

Detect Context Restarts - with Tooling

SPRING
TEST PROFILER

An open-source Spring Test utility that provides visualization and insights for Spring Test execution, with a focus on Spring context caching statistics.

Overall goal: Identify optimization opportunities in your Spring Test suite to speed up your builds and ship to production faster and with more confidence.

The Final Boss

Developers tend to consult AI/StackOverflow for integration test issues and often copy advice from the internet without knowing the implications:

```java
@SpringBootTest
@DirtiesContext
// this instructs Spring to remove the context from the cache
// and rebuild a new context on every request
public abstract class AbstractIntegrationTest {

}
```

The setup above will **disable** the context caching feature and slow down the builds significantly!

QUEST ITEM 2

The Lightning Shield

Many Cores, One Goal

Test Parallelization

Goal: Reduce build time and get faster feedback

Requirements:

- No shared state
- No dependency between tests and their execution order
- No mutation of global state

Two ways to achieve this:

- Fork a new JVM with Surefire/Failsafe (or for the Gradle test task) and let it run in parallel
- Use JUnit Jupiter's parallelization mode and let it run in the same JVM with multiple threads

Java Test Parallelization Options

Maven Surefire Fork-Based
(Multiple JVMs)

JVM Fork 1
- Test Class A
- Test Class B
- Test Class C

JVM Fork 2
- Test Class D
- Test Class E
- Test Class F

Maven (forkCount) Features:
- Each fork is a separate JVM process
- Isolated memory spaces
- Higher memory overhead
- High isolation (one DB per fork)
- Configuration: forkCount, reuseForks

JUnit Jupiter Thread-Based
(Single JVM)

Single JVM

Thread Pool

Thread 1	Thread 2
Test Method A1	Test Method A2
Test Method B1	Test Method B2
Test Class C	Test Class D

JUnit Jupiter Features:
- Uses threads within a single JVM
- Shared memory space
- Lower memory overhead
- Configuration: junit.jupiter.execution.parallel.*

Test Parallelization 101

Using Surefire/Failsafe:

```xml
<plugin>
  <artifactId>maven-surefire-plugin</artifactId>
  <configuration>
    <forkCount>1C</forkCount> <!-- 1 JVM per CPU core -->
  </configuration>
</plugin>
```

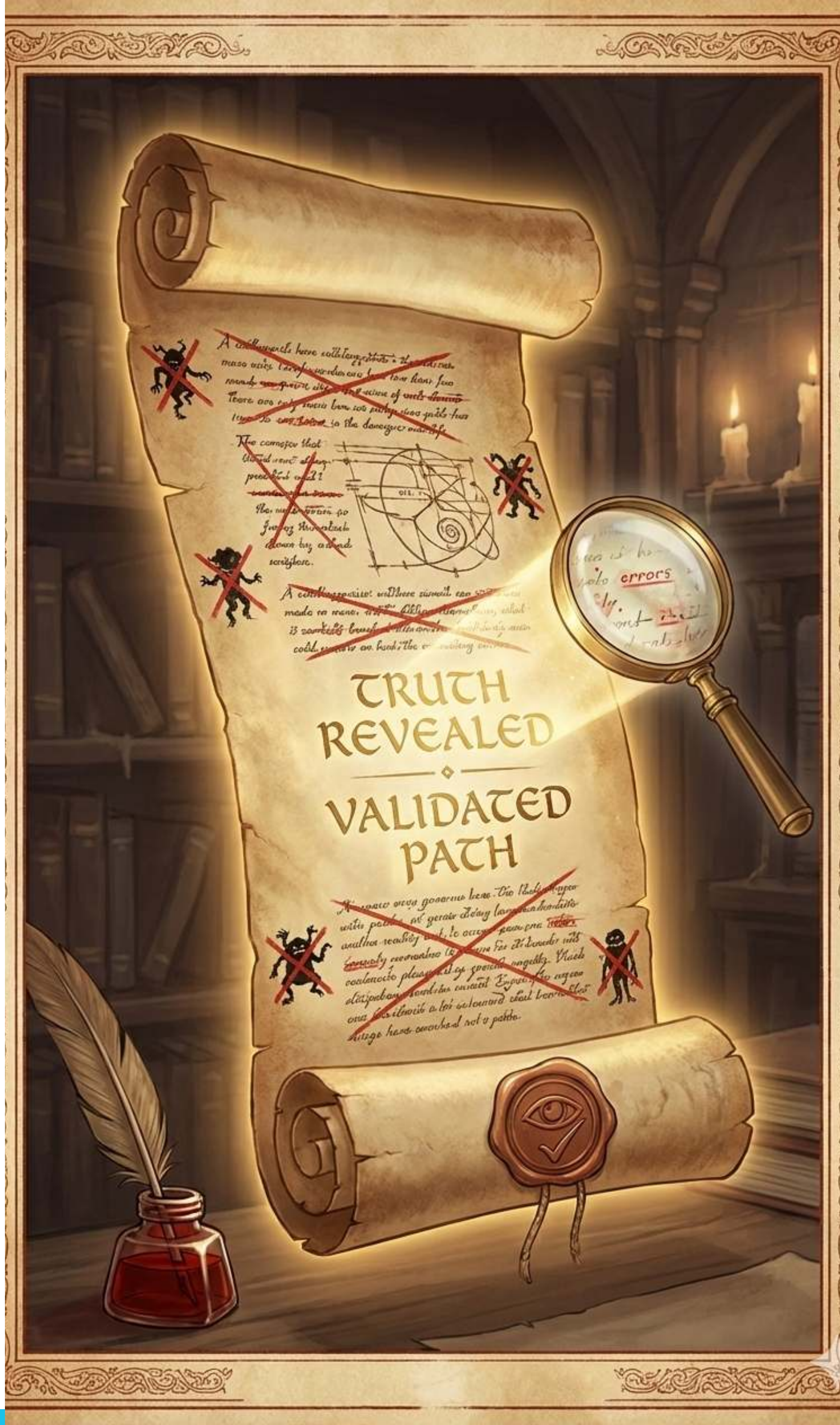
With JUnit Jupiter:

```properties
# src/test/resources/junit-platform.properties
junit.jupiter.execution.parallel.enabled = true
junit.jupiter.execution.parallel.mode.default = same_thread
junit.jupiter.execution.parallel.mode.classes.default = concurrent
```

QUEST ITEM 3

The Scroll of Truth

Coverage Lies, Mutants Don't



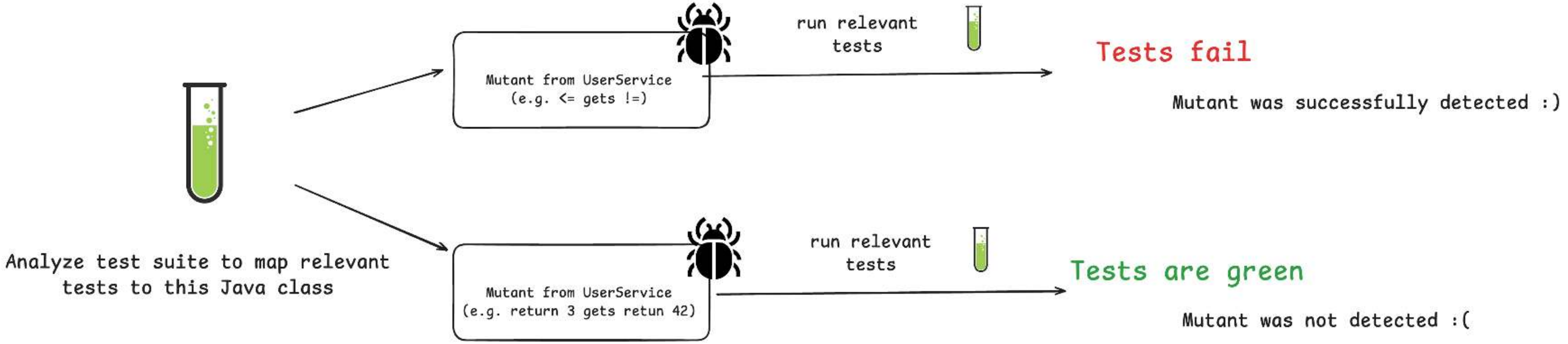
Let’s Challenge Code Coverage

Imagine a set of unit tests for this isolated business logic:

```java
public Long registerUser(int age, String username) {

  if (age <= 18) {
    throw new IllegalArgumentException("User must be at least 18 years old");
  }

  if ("ADMIN".equalsIgnoreCase(username)) {
    throw new IllegalArgumentException("Username 'ADMIN' is not allowed");
  }

  // ...

}
```

Idea: Introduce Regressions to Verify Test Quality

```java
if (age != 18) { // PIT mutated this with bytecode manipulation
  throw new IllegalArgumentException("User must be at least 18 years old");
}
```




```java
if ("ADMIN".equals(username)) {
  return 42L; // PIT mutated this with bytecode manipulation
}
```

Introducing: Mutation Testing

- Having high code coverage might give you a **false sense of security**
- Mutation Testing with PIT
- Beyond Line Coverage: Traditional tools like JaCoCo show which code runs during tests, but PIT verifies if our tests actually detect when code behaves incorrectly by introducing "**mutations**" to our source code.
- Quality Guarantee: PIT automatically **modifies our code** (changing conditionals, return values, etc.) to ensure our tests fail when they should, **revealing blind spots** in seemingly comprehensive test suites.

Act 5: The Triumphant Exit

- Spring Boot applications come with batteries-included and excellent testing support
- We've completed three main quests: Unit testing, Sliced testing, and Integration testing
- Three core quest items help us to speed up and validate our tests:
  - Context Caching
  - Test Parallelization
  - Mutation Testing

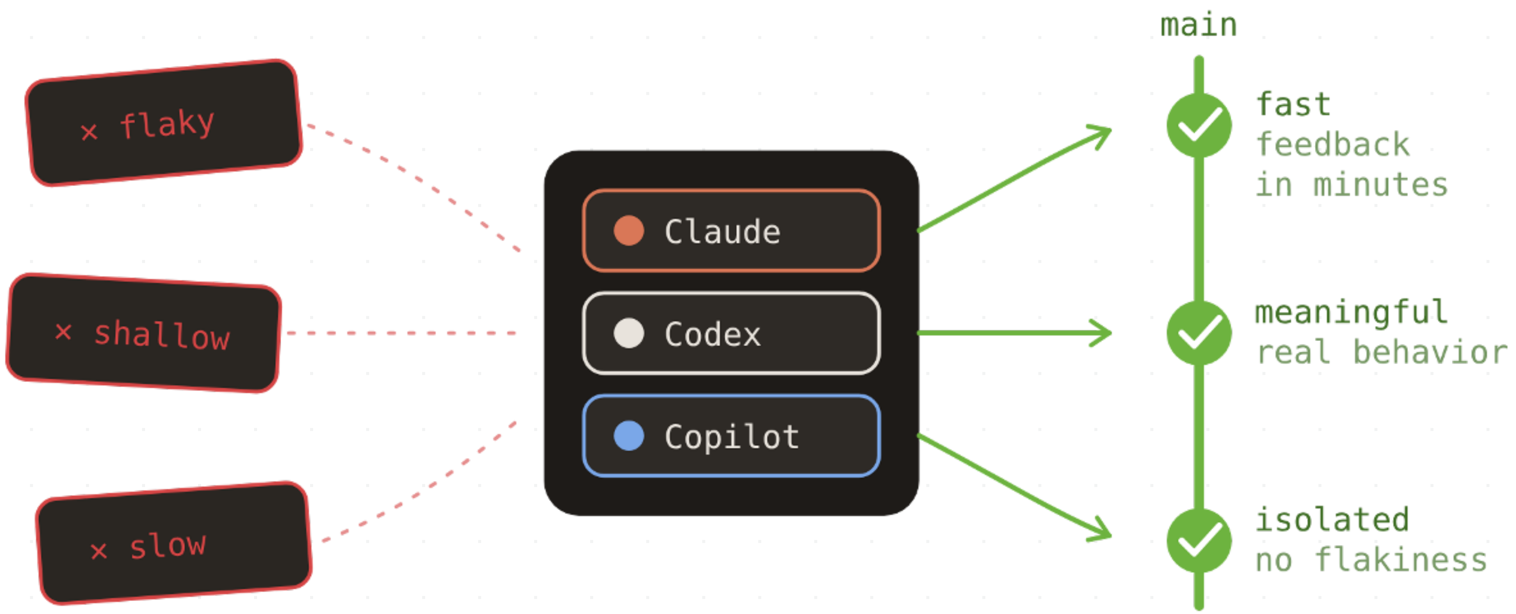
Spring Boot 4 - Testing Support Keeps Improving

- **RestTestClient**: Modern, fluent alternative for the `TestRestTemplate` / `WebTestClient` / `RestAssured`.
- **Context pausing**: Cached test contexts are now automatically paused, eliminating resource conflicts from background processes.
- **JUnit 6**: Drop-in upgrade from JUnit 5 - far smoother than the JUnit 4 → 5 migration.
- **Testcontainers 2.0**: New `testcontainers-` prefix for modules, JUnit 4 support removed.
- **Bean overrides for non-singletons**: `@MockitoBean` and `@TestBean` now work with prototype and custom-scoped beans.

Making the Most of Agentic Development

- Invest in a **comprehensive** and **fast test suite**
- Reduce context switches for developers to stay in the loop
- Achieve **operational excellence**: monitoring, alerting, runbooks, A/B testing, feature flags, canary testing etc.
- Use e.g. Playwright MCP to let the AI self-verify features locally
- Provide custom skills to standardize code and tests across the company

Upcoming Online Course



Agentic Testing for Spring Boot

From AI-generated noise to confidence in every commit

Optimizing AI-written Tests with Skills

Define a **skillset** for fast & comprehensive tests, including a **test strategy** for the given project:

```
.claude/skills/
├── unit-testing               fast tests without context bloat
├── slice-testing              right-sized Spring context slices
├── slice-testing-webmvc       web layer with @WebMvcTest
├── integration-testing        full-context tests that stay fast
├── e2e-testing                user journeys against the running app
├── testcontainers-setup       real infrastructure, reused containers
└── test-setup-reviewer        flags test anti-patterns for you
```

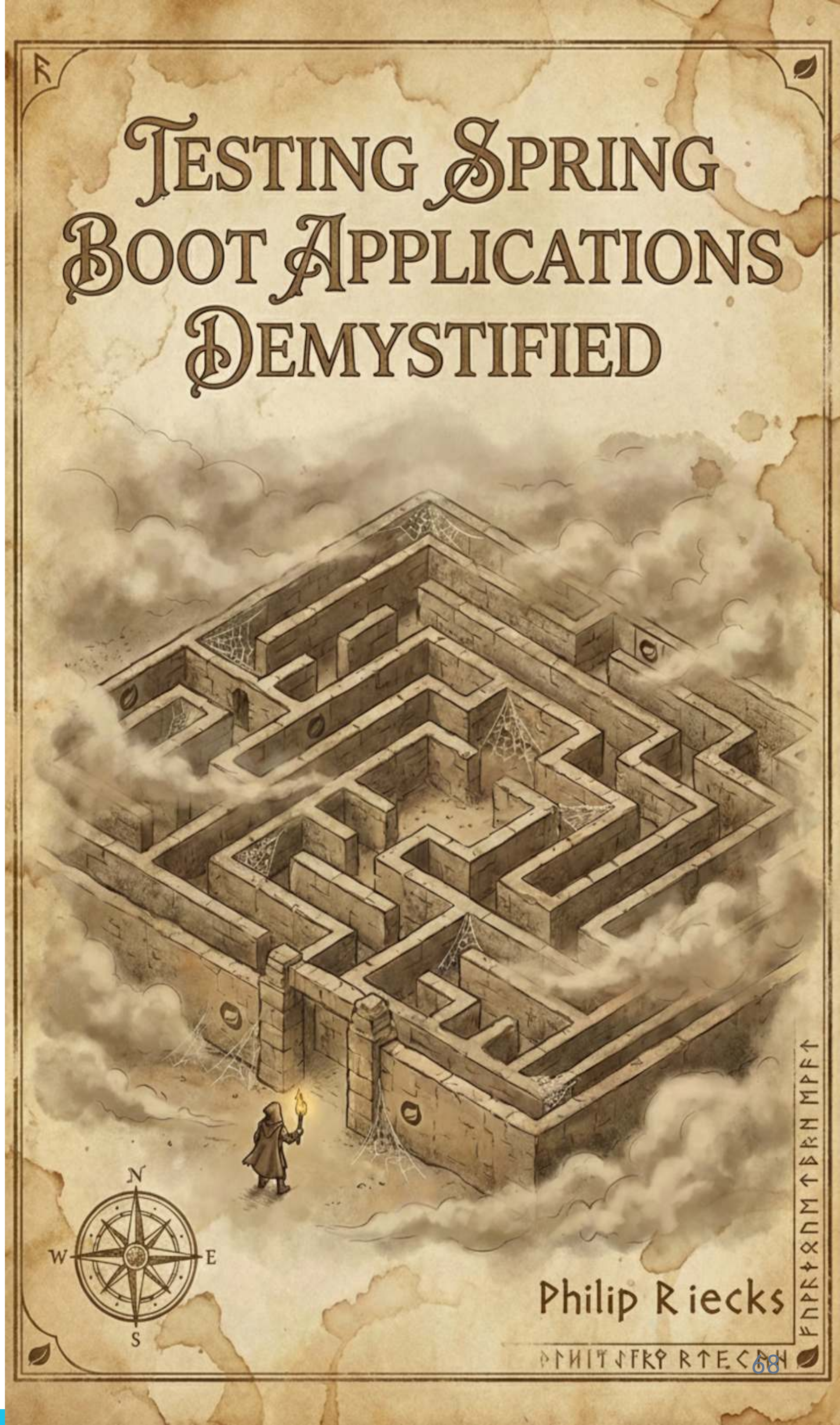
Each skill includes rules, references, best-practices and antipatterns described as code and text.

Let's Finalize the Menti Survey

Go to menti.com and use the code **3982 6427** to finalize the poll and add your questions for the Q&A.

Don’t Leave Empty-Handed

- Get the complementary **Testing Spring Boot Applications Demystified** for free (instead of $9)
- 120+ Pages with practical hands-on advice to ship code with confidence
- Get the eBook by joining our newsletter



Joyful Testing!

Get your Spring Boot Testing eBook (120+ pages):

Reach out any time

- LinkedIn (Philip Riecks)
- Mail (philip@pragmatech.digital)